HANY EL ATLASSI

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Profile

AI & Cloud engineer with hands-on experience building and deploying ML systems on AWS: voice pipelines, multi-agent orchestration, federated learning, and the security infrastructure underneath all of it, ensuring complete deployment. Available and open to relocation.

Work Experience

Alignerr | ***AI Trainer – Code Specialist*** — Oct 2024 – Present

- Evaluate AI-generated code for logic correctness, edge cases, and optimization across multiple model families, output feeds RLHF pipelines for production LLMs
- Assess agentic behavior and function-calling decisions in multi-step workflows, focus on failure modes in tool chaining, hallucinated API calls, and malformed structured output

3D Smart Factory | ***Machine Learning Intern*** — Jun 2024 – Sep 2024

- Trained Superpoint Transformer on **30 GB+** Stanford 3D Indoor Scenes, reaching **90%** accuracy and **70%** mean IoU on 3D point cloud segmentation; end-to-end from raw data preprocessing to a deployed real-time Streamlit demo

CIAM AI Club, ENSAM | ***President*** (**200+** members) — Sep 2023 – Jun 2025

- Grew and ran ENSAM's AI community to **200+** members; organised weekly technical workshops, and coordinated student participation in national AI competitions

Education

École Nationale Supérieure d'Arts et Métiers (ENSAM Casablanca) — 2022 – 2027

Engineer's Degree — Cybersecurity & Cloud Engineering — ***GPA:*** *3.7/4.0*

- Research paper in preparation: FedLoRA-MAML, federated second-order meta-learning for personalised accented speech recognition (Wav2Vec2, LoRA, Per-FedAvg, L2-ARCTIC corpus)

Projects

InterviewForge — Voice AI Interview Platform — 2026 – Present

- Designing and building a **5-layer concurrent voice architecture**: Deepgram Flux + Pipecat Smart Turn v3 (**12 ms** CPU) → gpt-4o-realtime-mini filler at ∼**329 ms** → Cartesia Sonic-3 (**40–90 ms** TTFB) over LiveKit WebRTC.
- **7-agent LangGraph** pipeline; knowledge grounded via Claude cached prefix (**90%** cost reduction vs RAG); Judge0 + gVisor sandboxed code execution (seccomp, cgroups, zero network egress); K8s deployment behind Kong and Cloudflare WAF, GitOps via ArgoCD

VoiceFL-MAML — Federated Meta-Learning for Accented Speech | github — 2025 – 2026

- Built FedLoRA-MAML across **12 distributed nodes** spanning **6** non-native English accent groups (L2-ARCTIC); at round 50, the federated meta-initialisation matches or outperforms wav2vec2-base-100h, including a statistically significant −**3.1 pp** WER improvement on Vietnamese ($p < 0.05$, non-overlapping 95% CI)
- Cut per-round communication ∼**300**× by transmitting only **320K** LoRA parameters (rank-8 adapters on transformer layers 6–11) instead of 94.5M full encoder weights
- Engineered a custom differentiable CTC loss in native PyTorch to unlock true second-order MAML; `nn.CTCLoss` has no registered second derivative, this was the part that made the math actually work

Achievements

- **1st Place, AgorAI Hackathon** — won with Aegis, an AI policy impact simulator for Moroccan AI regulation; presented at UM6P Spring School AI for Impact alongside Yann LeCun, Eric Xing, and researchers from Google DeepMind, UC Berkeley, and universities worldwide
- **2nd Place, HackAI 5th Edition (1337AI)** — won with best pitch award, building a voice-first AI agronomist for Moroccan farmers; full farming plan orchestration, hallucination-hardened via a 5-layer safety pipeline, responding in Darija
- **Top 10 globally, Mistral AI Worldwide Hackathon** — MediCore, AI clinical safety assistant built on Mistral OCR + a real-time voice pipeline
- **CTF:** Top **3%** worldwide on HackTheBox (Forensics track, 600K+ registered users); top **10–15** nationally across multiple Moroccan CTF competitions

Skills & Certifications

- **AI/ML:** PyTorch, TensorFlow, Scikit-Learn, LangGraph, LangChain, Flower (Federated Learning), Opacus, LiveKit, Deepgram, Cartesia
- **Cloud & DevOps:** AWS (EKS, SageMaker, Lambda, ECR), Kubernetes, Docker, Terraform, Kong, ArgoCD, Git/GitHub, Linux
- **Security:** OS/memory forensics, network analysis, NTFS artifacts, prompt injection hardening
- **Languages:** Python, C/C++, Java, SQL, Bash | English (Fluent), French (Fluent), Arabic (Native)
- **Certifications:** IBM AI Engineering, Deep Learning and Reinforcement Learning Specializations